

A PROJECT TITLE

ON

Malicious Traffic Pattern Clustering Using K-Means and Hierarchical Clustering
project report submitted in partial fulfilment of the requirements for the Course

MACHINE LEARNING

BACHELOR OF TECHNOLOGY

in

A.Y.: 2026-27

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DEPARTMENT OF AI&DS

Name of Title: Malicious Traffic Pattern Clustering Using K-Means and Hierarchical Clustering

1. Abstract:

The rapid growth of internet usage has led to a significant increase in network traffic, making it more difficult to identify malicious activities and maintain network security. Traditional security methods are effective in detecting known threats, but they may not always identify new or unusual patterns of network behaviour. Since most real-world network traffic is not labelled, analysing it manually is both time-consuming and challenging. This project addresses this problem by using unsupervised machine learning to group similar network traffic based on its characteristics instead of relying on predefined labels. The proposed system uses K-Means and Hierarchical Clustering algorithms to analyse network traffic using features such as connection duration, packet count, byte count, protocol type, and destination port. The dataset is first cleaned and pre-processed by handling missing values, removing unnecessary information, and scaling the selected features to improve the clustering process. Both clustering techniques are then applied to group traffic with similar behaviour, allowing patterns and possible anomalies to be identified. The clustered data is further processed using Principal Component Analysis (PCA) to reduce the number of dimensions and make the results easier to visualize and interpret. To improve the understanding of the clustered data, the project includes an interactive visualization dashboard that displays the generated clusters in a clear and user-friendly manner. This helps users observe traffic patterns, compare different clusters, and identify groups that may represent suspicious network behaviour. The outcome of this project demonstrates how clustering techniques can be used to analyse large volumes of network traffic without requiring labelled datasets. By comparing the results of K-Means and Hierarchical Clustering and presenting them through meaningful visualizations, the project provides a simple and practical approach for network traffic analysis and supports the early identification of unusual or potentially malicious activities.

2. INTRODUCTION:

The rapid growth of internet-connected devices has significantly increased the volume and complexity of network traffic, making it difficult to identify malicious activities using traditional methods alone. Unsupervised machine learning offers an effective approach to discovering hidden patterns in large, unlabelled datasets without requiring prior knowledge of attack types. This project applies K-Means and Hierarchical Clustering algorithms to group similar network traffic patterns, enabling the identification and analysis of potentially malicious behaviour. The results are further visualized to provide meaningful insights into network traffic characteristics and support cybersecurity analysis.

3. SOFTWARE REQUIREMENTS:

Python 3.10 or above

Jupyter Notebook / Google Colab

Pandas

NumPy

Scikit-learn

SciPy

Matplotlib

Plotly

VS Code / PyCharm

Git & GitHub



Signature of the Guide



Course Instructor